Navid Aftabi

Department of Industrial and Systems Engineering
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Education

Ph.D. in Industrial and Systems Engineering (Minor: Computer Science) *2022 – Exp. June 2027*
University of Washington Seattle, WA, USA

- Dissertation: *Toward Resilient Cyber-Physical Systems: From Strategic Attack Modeling to Adaptive Defense*
- Advisor: Prof. Dan Li (chair). Committee: Profs. Satish T.S. Bukkapatnam, Shuai Huang, Thomas C. Sharkey, Krithika Manohar

M.Sc. in Industrial Engineering *2020 – 2022*
Sabanci University Istanbul, Turkey

M.Sc. in Industrial and Systems Engineering *2016 – 2019*
Sharif University of Technology Tehran, Iran

B.Sc. in Computer Engineering *2010 – 2015*
University of Tabriz Tabriz, Iran

Professional Experience

Graduate Research Assistant *August 2022 – Present*
Department of Industrial & Systems Engineering, University of Washington, Seattle, WA *June 2025 – Present*
Department of Industrial & Systems Engineering, University of Wisconsin–Madison, WI *May 2024 – June 2025*
Department of Industrial Engineering, Clemson University, Clemson, SC *August 2022 – May 2024*

- Continuous appointment in the same research group; the lab relocated with the advisor.

Machine Learning Engineering Ph.D. Intern *June 2026 – September 2026*
Instacart

Research and Teaching Assistant *January 2020 – August 2022*
Department of Industrial Engineering, Sabanci University, Istanbul, Turkey

Research Assistant *January 2017 – January 2019*
Department of Industrial Engineering, Sharif University of Technology, Tehran, Iran

Research Interests

Cyber-Physical Security & Resilience • Online Anomaly & Cyberattack Detection • Cyberattack-Resilient Control • Uncertainty Quantification • Reinforcement Learning, MDPs & POMDPs • Physics-Informed & High-Dimensional Machine Learning • Multivariate & Bayesian Statistics • Stochastic Systems Modeling • Applications: Smart Manufacturing, Power, Energy, Transportation, and Healthcare Systems, Supply Chains.

Honors and Awards

- **IISE Transactions Featured Paper (Top 4) [J.4]** *2026*
  Selected by the Editor-in-Chief of IISE Transactions for the journal's featured session at 2026 ICQSR, Hong Kong
- **IISE DAIS Best Student Paper Award (Finalist) [J.6]** *2025*
  Data Analytics and Information Systems (DAIS) Division, Institute of Industrial and Systems Engineers (IISE) [🌍]
- **Industrial Engineering Travel Award** *2023*
  Industrial Engineering Department, Clemson University
- **GTG Travel Award** *2023*
  Graduate Student Government, Clemson University
- **Top 0.08% (11/14,000), National M.Sc. Entrance Exam, Industrial Engineering** *2016*
  Nationwide ranking among ~14,000 candidates

Publications

J=Journal, C=Conference, S=Submitted, W=Working Paper, PT=Patent

Google Scholar: 140 citations (as of October 2026)

Refereed Journal Papers (Published / Accepted)

[J.6] **Navid Aftabi**, Abhishek Hanchate, Satish T.S. Bukkapatnam, Dan Li (2026). *"DynaMark: A Reinforcement Learning Framework for Dynamic Watermarking in Industrial Machine Tool Controllers"*. **IEEE Transactions on Automation Science and Engineering**, Vol. 23, pp. 13658–13676. DOI: 10.1109/TASE.2026.3715642.
🏆 *IISE DAIS Best Student Paper Award (Finalist), 2025*

[J.5] **Navid Aftabi**, Philip Samaha, Jin Ma, Long Cheng, Ramy Harik, Dan Li (2026). *"ViSTR-GP: Online Cyberattack Detection via Vision-to-State Tensor Regression and Gaussian Processes in Automated Robotic Operations"*. **Journal of Manufacturing Systems**, Vol. 89, pp. 504–523. DOI: 10.1016/j.jmsy.2026.09.008.

[J.4] **Navid Aftabi**, Dan Li, Thomas C. Sharkey (2025). *"An Integrated Cyber-Physical Framework for Worst-Case Attacks in Industrial Control Systems"*. **IISE Transactions**, Vol. 57, Issue 12, pp. 1464–1482. DOI: 10.1080/24725854.2024.2439856.

🏆 *Top-4 IISE Transactions Featured Paper, 2026 ICQSR, Hong Kong*

[J.3] Fatemeh Mahroo, Nima Moradi, **Navid Aftabi**, Mahmoud Houshmand, Omid Fatahi Valilai (2026). *"A Novel Integer Linear Model for Reliability-Centric Service Composition in Cloud Manufacturing"*. **Computers & Industrial Engineering**, Vol. 211, pp. 111603. DOI: 10.1016/j.cie.2025.111603.

[J.2] **Navid Aftabi**, Nima Moradi, Fatemeh Mahroo, Farhad Kianfar (2025). *"SD-ABM-ISM: An Integrated System Dynamics and Agent-based Modeling Framework for Information Security Management in Complex Information Systems with Multi-Actor Threat Dynamics"*. **Expert Systems with Applications**, Vol. 263, pp. 125681. DOI: 10.1016/j.eswa.2024.125681.

[J.1] **Navid Aftabi**, Nima Moradi, Fatemeh Mahroo (2025). *"Feed-Forward Neural Networks as a Mixed-Integer Program"*. **Engineering with Computers**, Vol. 41, pp. 2411–2429. DOI: 10.1007/s00366-025-02114-2.

Refereed Conference Papers (Published / Accepted)

[C.1] **Navid Aftabi**, Paritosh Ramanan, Dan Li (2026). *"PINPOINT: Probabilistic Residual Fusion for Cyberattack Detection and Localization in Power Systems"*. Manuscript accepted for publication in **25th IEEE International Conference on Machine Learning and Applications (ICMLA'26)**.

Submitted Papers (Under Review / Revision)

[S.2] Aditya Parameshwaran, **Navid Aftabi**, Abhishek Hanchate, Satish Bukkapatnam, Dan Li, Yue Wang (2026+). *"Dynamic Watermarking and Action-Conditioned Latent Dynamics for Secure Vision-Based Controllers"*. Manuscript under review for publication in **2027 American Control Conference (ACC)**.

[S.1] Nima Moradi, Niloufar Mirzavand Boroujeni, **Navid Aftabi**, Amin Aslani (2024+). *"Two-echelon Electric Vehicle Routing Problem in Parcel Delivery: A Literature Review"*. Manuscript *under revision* for publication in **European Journal of Operational Research**.

Working Papers

[W.3] **Navid Aftabi**, Dan Li. *"SUBVERT: A Stealth-Constrained Strategic Cyberattack in Unified Cyber-Physical Systems via Reinforcement Learning"*, working paper, target journal: **IISE Transactions**.

[W.2] **Navid Aftabi**, Philip Samaha, Ramy Harik, Dan Li. *"ViST-DiFA: Online Vision-to-State Diagnosis for Differentiating Faults and Cyberattacks in Automated Robotic Operations"*, under preparation, target journal: **Journal of Manufacturing Systems**.

[W.1] **Navid Aftabi**, Dan Li, Prashanth Rajivan. *"CyberReady: Uncertainty-Aware Personalized Anti-Phishing Training under Sparse Behavioral Feedback and Limited Training Budgets"*, working paper.

Patents

[PT.1] **Navid Aftabi**, Zi Wang, Houtao Deng. *"A Personalized & Targeted Sequential Retention Incentivization System"*, patent in preparation (Instacart).

Teaching Experience

Teaching Assistant

- **ENGR 520 A: Physics-Informed Machine Learning (Graduate)** — *Spring 2026*
  University of Washington
- **IE 305: Simulation (Undergraduate)** — *Spring 2020, 2021; Fall 2021*
  Sabanci University
- **MATH 306: Statistical Modeling (Undergraduate)** — *Spring 2022*
  Sabanci University
- **ENS 208 A,B: Introduction to Industrial Engineering (Undergraduate)** — *Fall 2020*
  Sabanci University

Teaching Training

- Teaching@UW: Strategies for TAs, University of Washington, 2026.
- Grad WAC Fellow (Writing Across the Curriculum), Pearce Center, Clemson University, 2023–2024.
- Teaching Assistant Training and Orientation, Sabanci University, 2020.

Mentoring Experience

- **Mentor, NSF Future Manufacturing Data Challenge** — *2026*
  Texas A&M University [🌍]

- **Research Mentor, Undergraduate Researchers (Creative Inquiry Program)** *2022 – 2024*
  Clemson University
- **Research Mentor, M.Sc. Student** *2022 – 2024*
  Clemson University

Selected Presentations

SUBVERT: A Stealth-Constrained Strategic Cyberattack in Unified Cyber-Physical Systems via Reinforcement Learning *2026*
IISE Annual Conference Arlington, TX, USA

DynaMark: A Reinforcement Learning Framework for Dynamic Watermarking in Industrial Machine Tool Controllers *2025*
International Mechanical Engineering Congress & Exposition (IMECE) Memphis, TN, USA

SUBVERT: A Stealth-Constrained Strategic Cyberattack in Unified Cyber-Physical Systems via Reinforcement Learning *2025*
INFORMS Annual Conference Atlanta, GA, USA

DynaMark: A Reinforcement Learning Framework for Dynamic Watermarking in Industrial Machine Tool Controllers *2025*
IISE Annual Conference Atlanta, GA, USA

DynaMark: A Reinforcement Learning Framework for Dynamic Watermarking in Industrial Machine Tool Controllers *2024*
INFORMS Annual Conference Seattle, WA, USA

A Case Study on Optimal Watermarking in Machine Tool Controllers *2024*
International Mechanical Engineering Congress & Exposition (IMECE) Portland, OR, USA

An Integrated Cyber-Physical Risk Assessment Framework for Worst-Case Attacks in Industrial Control Systems *2023*
INFORMS Annual Conference Phoenix, AZ, USA

An Optimization Framework for Cyber-Physical Vulnerability Analysis in Industrial Cyber-Physical Systems *2023*
IISE Annual Conference New Orleans, LA, USA

Deep Learning Algorithms For Learning the Objective Value of the VRP *2021*
INFORMS Annual Conference Anaheim, CA, USA

Professional Service

Track Chair

• Cybersecurity Track (Co-Chair with Dr. Lynette Wilcox), IISE 2026, Arlington, TX, USA.

Session Chair

• Data Science for Cyber-Physical Security and Resilience, INFORMS 2026, San Francisco, CA, USA.

• Under Attack: Ransomware and Cyber Attack Modeling for Industrial Environments, IISE 2026, Arlington, TX, USA.

• Data Science for Cyber-Physical Security and Resilience, INFORMS 2025, Atlanta, GA, USA.

• Data Science for Cyber-Physical Security and Resilience, INFORMS 2024, Seattle, WA, USA.

• Deep Learning/Machine Learning II, INFORMS 2021, Anaheim, CA, USA.

Journal Referee

• IEEE Transactions on Industrial Informatics • IEEE Transactions on Automation Science and Engineering • IEEE Transactions on Cybernetics • IEEE Transactions on Engineering Management • IEEE Transactions on Control of Network Systems • Reliability Engineering & System Safety • ASME Journal of Computing and Information Science in Engineering • Nonlinear Dynamics • Expert Systems with Applications • The Journal of Supercomputing • Signal, Image and Video Processing • Machine Learning with Applications

Conference Referee

• IISE Annual Conference and Expo • SME's North American Manufacturing Research Conference (NAMRC) • ASME International Mechanical Engineering Congress & Exposition (IMECE) • IEEE International Conference on Intelligent Transportation Systems • International Conference on Machine Learning and Applications (ICMLA) • American Control Conference (ACC)

Professional Memberships

- **Institute for Operations Research and the Management Sciences (INFORMS)** *2021 – Present*
- **Institute of Industrial & Systems Engineers (IISE)** *2023 – Present*
- **Society of Manufacturing Engineers (SME)** *2024 – Present*